QUANTITATIVE FINANCE · ERDŐS INSTITUTE · FALL 2026

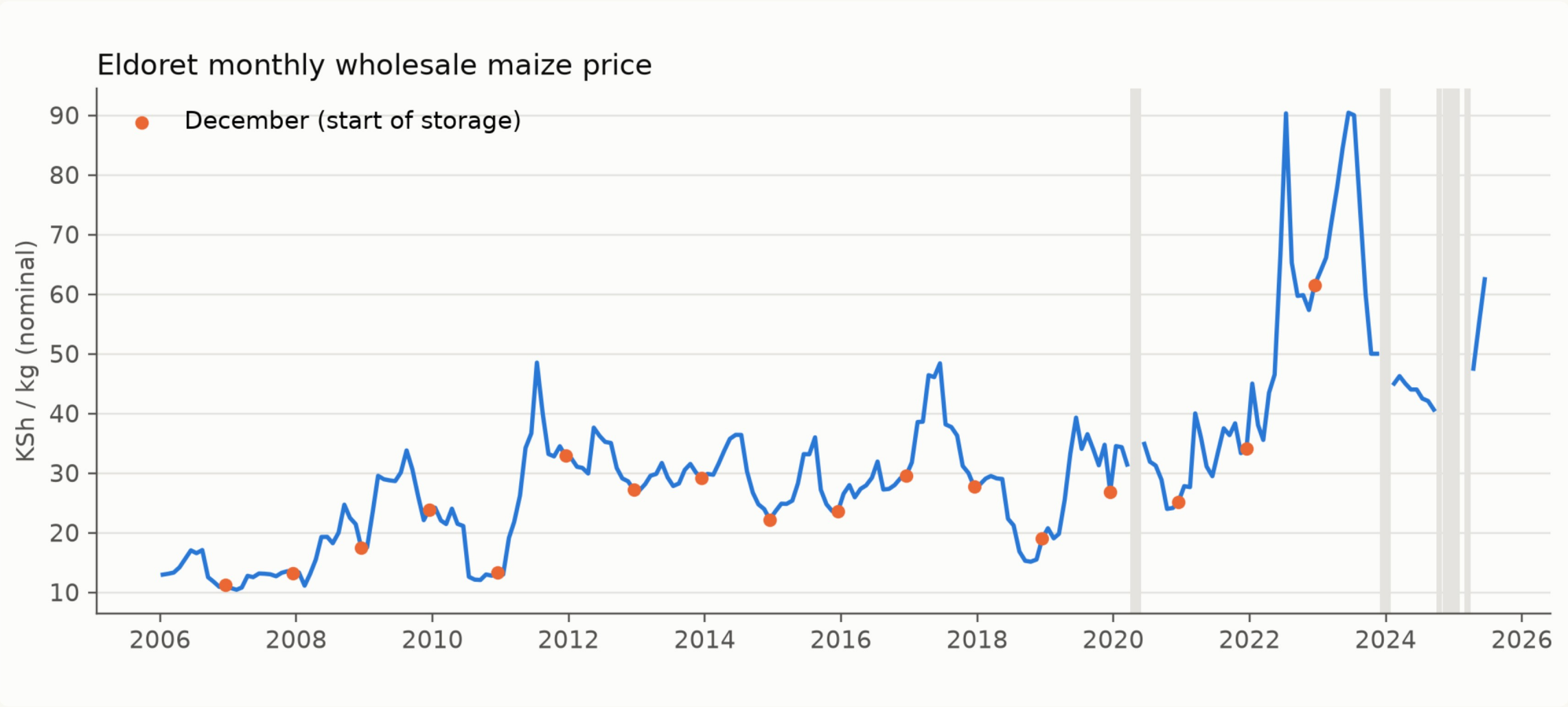
The optimal time to sell maize (Uasin-Gishu, Kenya)

Inspired by the Zindi agriBORA Commodity Price Forecasting Challenge

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THE DATA

19 years of monthly maize prices



Data: WFP Kenya Food Prices (HDX, CC BY-IGO); KAMIS via the Zindi agriBORA challenge

THE IDEA IN ONE INEQUALITY

Quantifying costs and benefits of waiting

$$E[\,P_t / P_0\,] \times e^{-(i + c)\cdot t/12} > 1$$

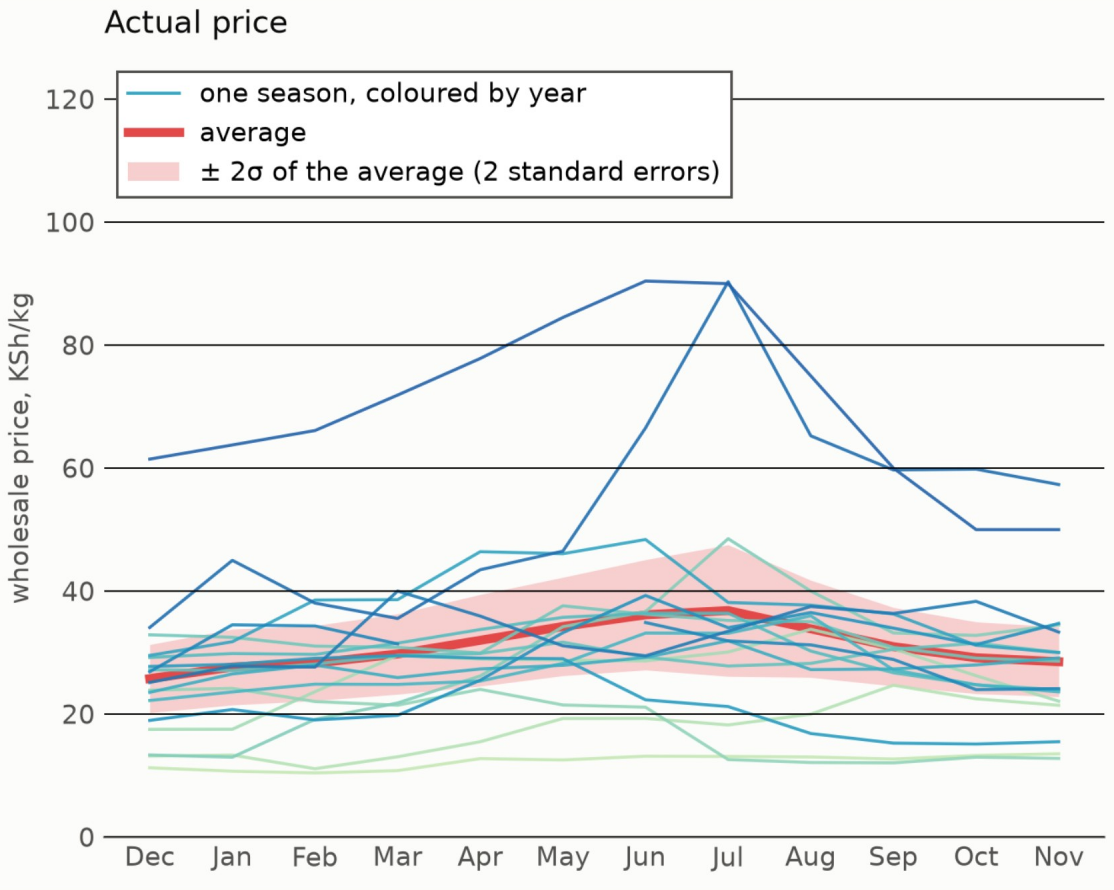
Symbol	Meaning	Value used
t, P0, Pt	months after December; December price; price at sale	sale at end of Jan … Nov
i (interest)	cost of the money you wait for (loan rate)	15% / yr
c (holding cost)	storage fee + losses	warehouse 3% / yr (+1.5% once); on-farm 20% / yr losses

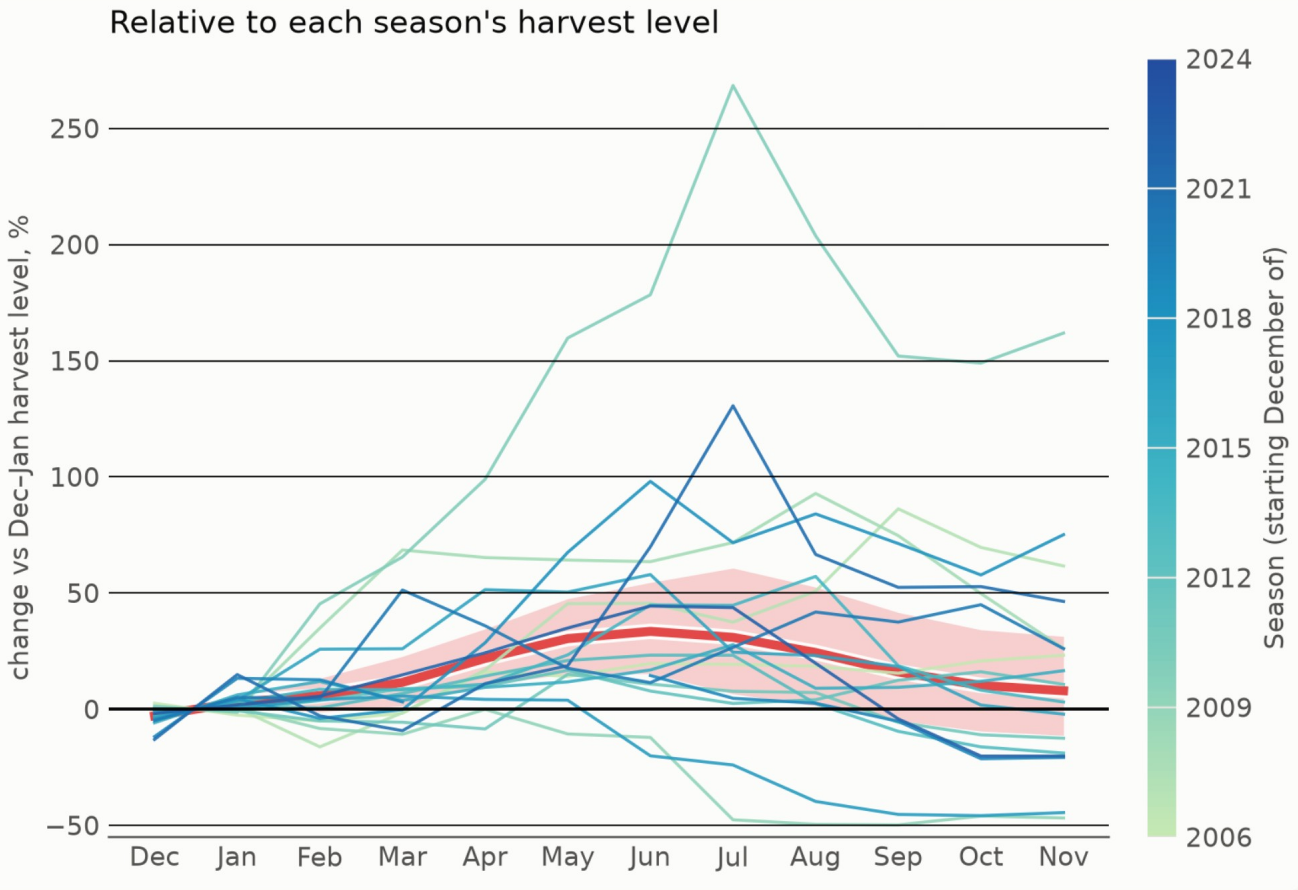
Interest: Central Bank of Kenya average lending rate (15.4% May 2025; 14.8% Dec 2025). Fees: Warehouse Receipt System tariff. Losses: De Groote et al. (2023)

Almost every year, prices peak in June

15 / 17 seasons higher in May–Jul than at harvest (sign test p ≈ 0.001)

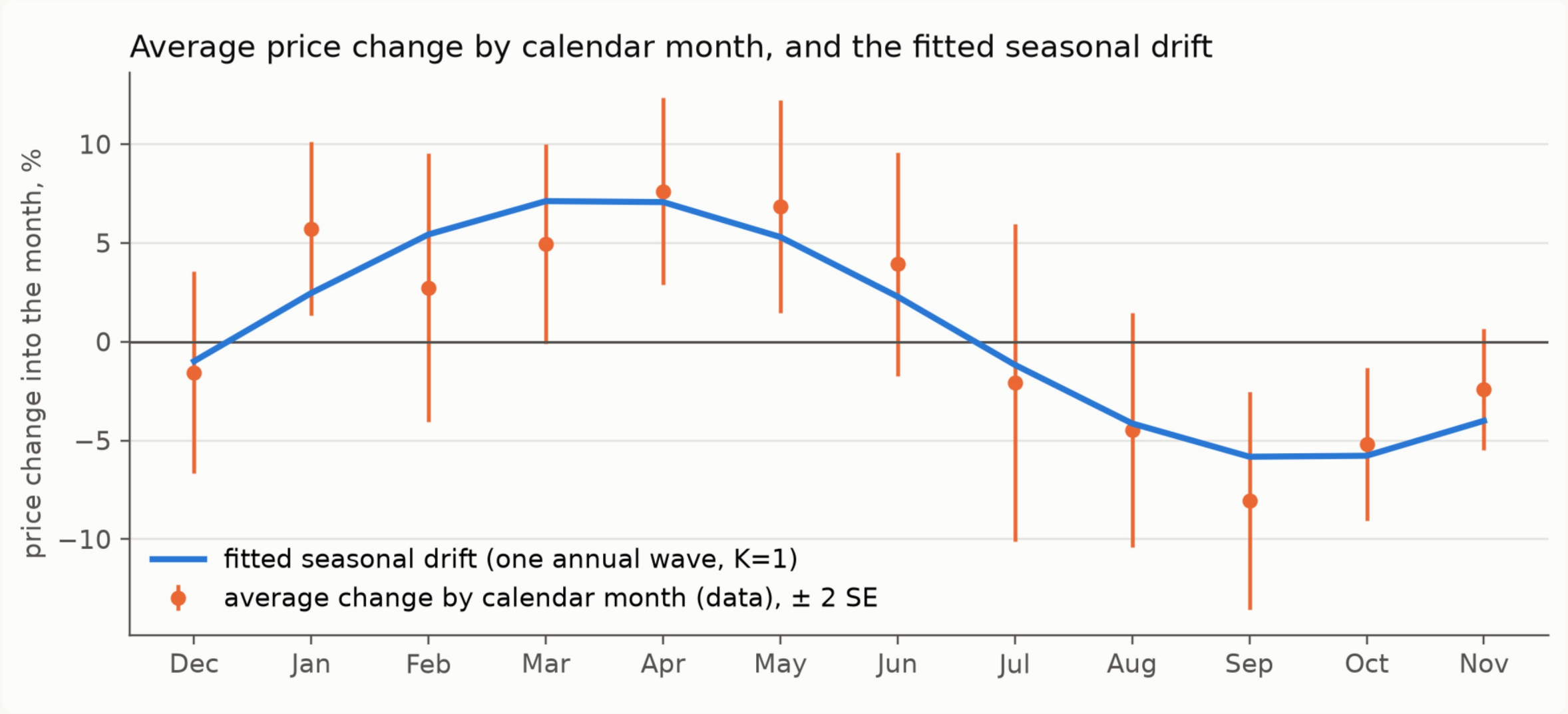
p ≈ 0.0002 annual cycle, random-phase permutation test





One sine wave captures the cycle

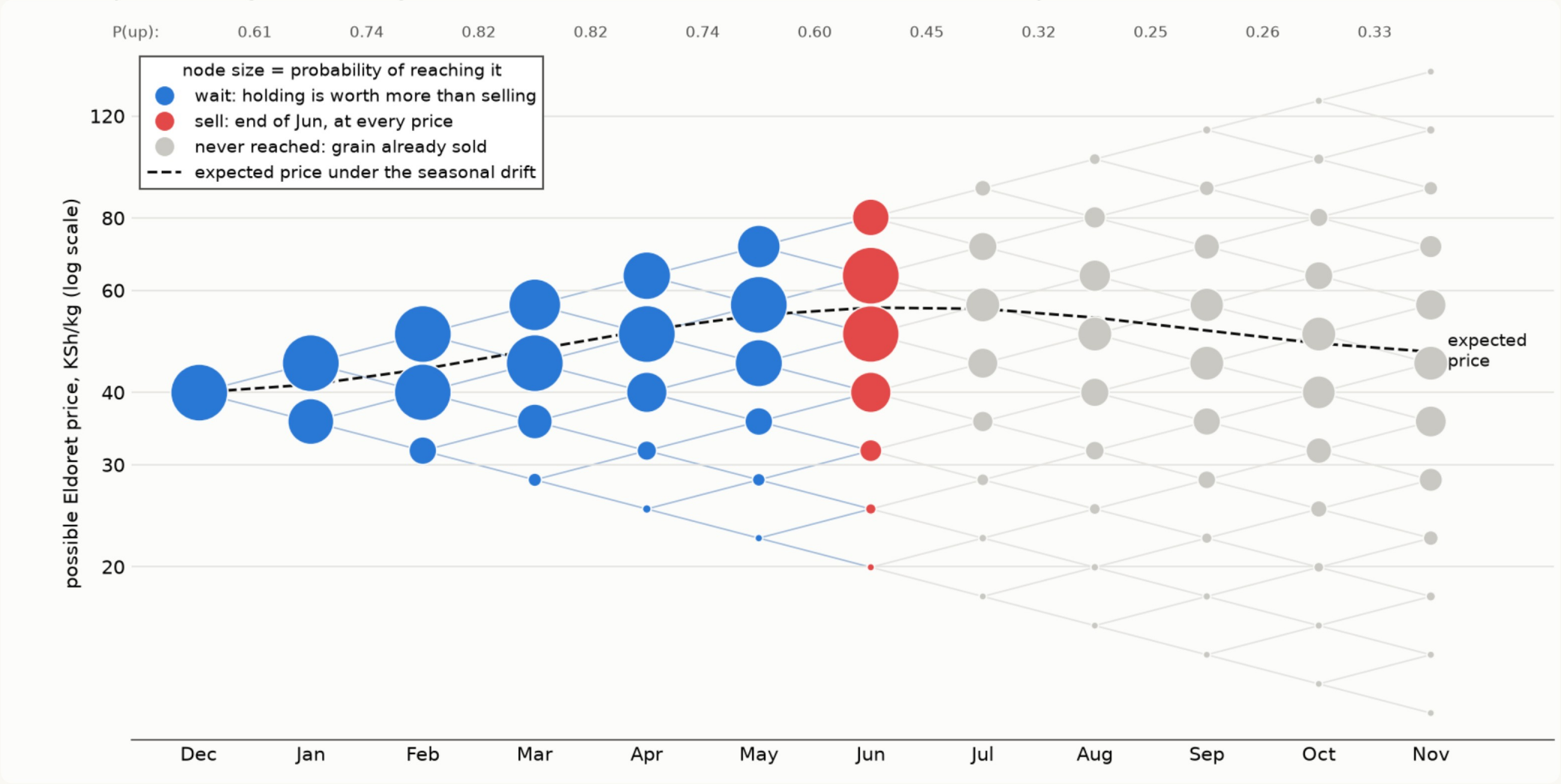
Fastest rise in Mar–Apr;
price in June, ≈ **+34%** above December



Points: average log return into each month ± 2 SE (17–19 years each). Line: least-squares fit

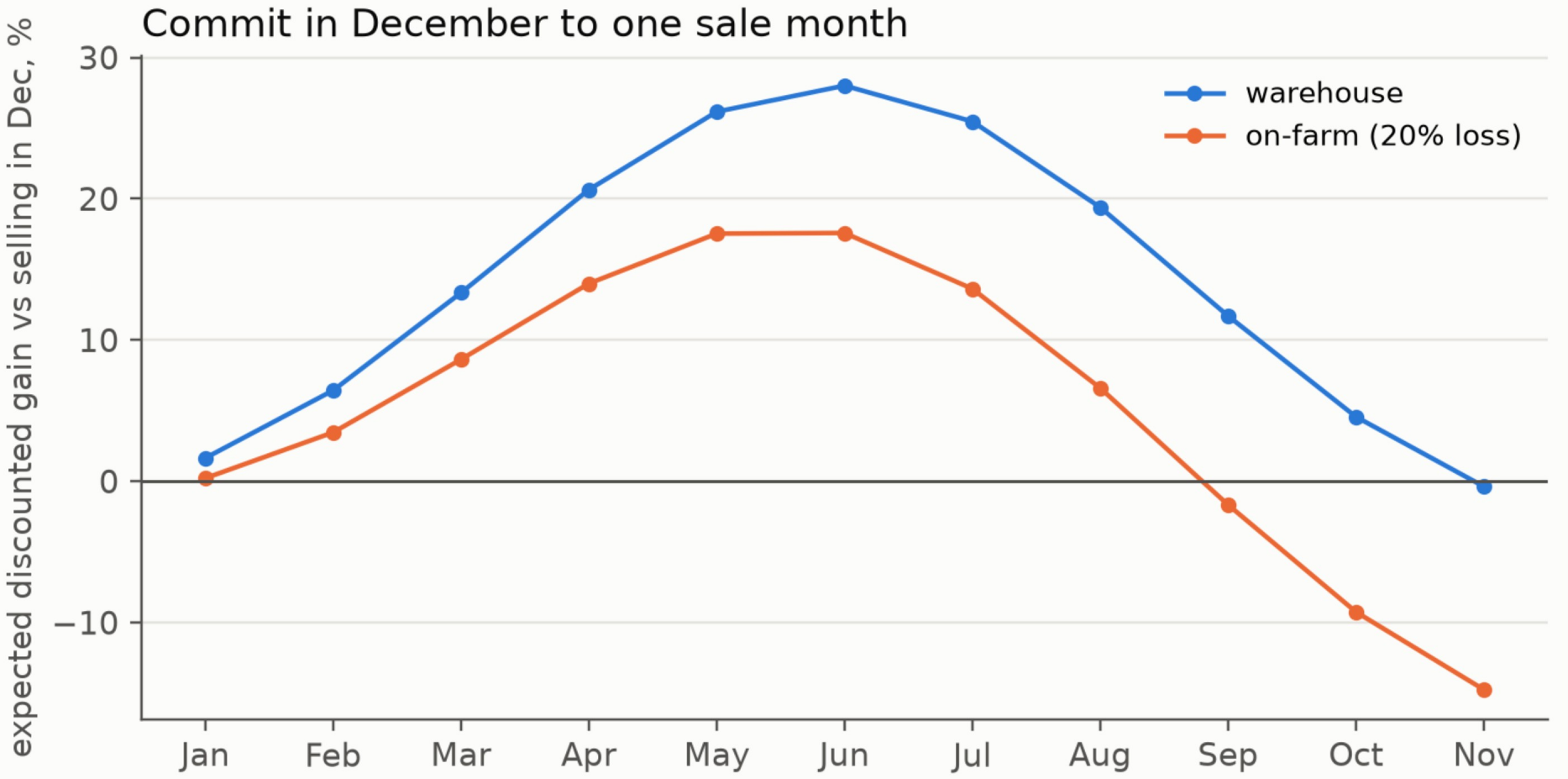
Optimal stopping calculated as an American-style option

Expected growth depends only on the month and all costs scale with price, so value = P · f(t): **only the date matters**



Warehouse case, one step per month. Blue: wait · red: sell · grey: already sold. Node size = probability of reaching it

Holding until June beats selling at harvest



+27%

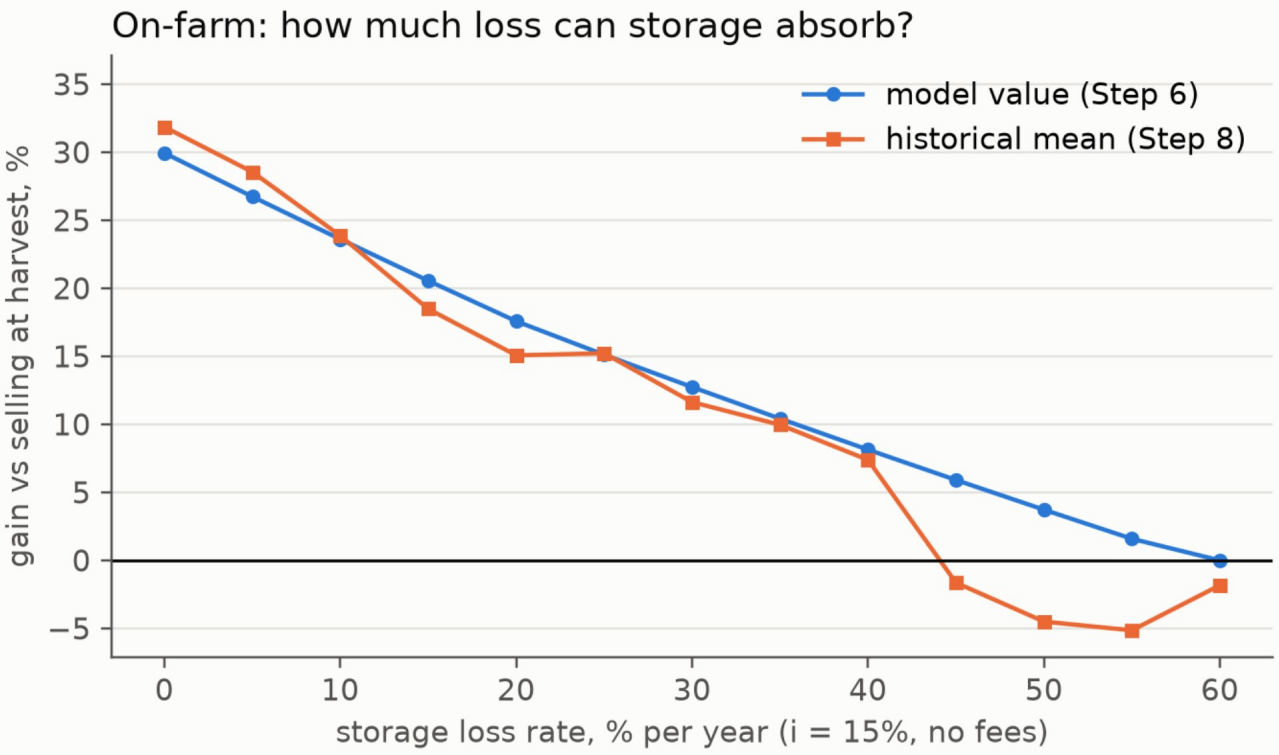
certified warehouse (≈ KSh 950 per 90-kg bag at 40 KSh/kg)

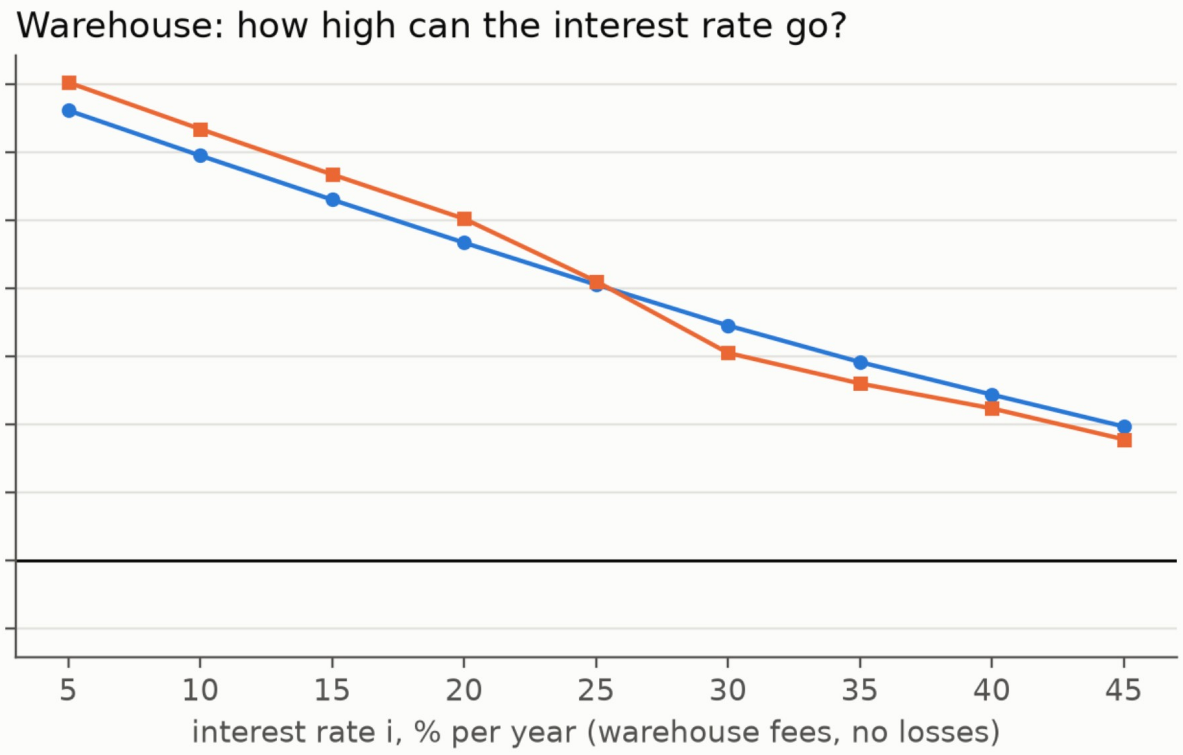
+18%

on the farm, with 20% a year of grain lost

Model value net of 15% interest, fees and losses, per KSh of the December price. +34% typical rise → +40% expected → +28% after carry → +27% after the one-off fee

Storage keeps paying until costs are far above today's





Warehouse: pays up to ≈ 53–68% interest (today ≈ 15%)

On-farm: pays up to ≈ 45–60% losses a year

So: a warehouse's value is removing losses

Ranges: historical average (lower) to model value (higher). Break-even = where the gain reaches zero

SOURCES

Data and references

Price data

World Food Programme (2026). *Kenya – Food Prices* [data set]. Humanitarian Data Exchange, CC BY-IGO. data.humdata.org/dataset/wfp-food-prices-for-kenya

Zindi & agriBORA (2025–26). *agriBORA Commodity Price Forecasting Challenge* [data set; KAMIS, Kenya Ministry of Agriculture]. zindi.africa

Code

github.com/thummim-mekuria/erdos_project — notebook, data pipeline, 38 unit tests

Cost inputs

Kenduiywo, B. et al. (2026). *Lessons from the agriGHALA warehouse receipt model in Kenya*. CIAT / agriBORA / P4G policy brief. hdl.handle.net/10568/182900

De Groote, H., Muteti, F. N. & Bruce, A. Y. (2023). On-farm storage loss estimates of maize in Kenya. *J. Stored Products Research* 102, 102107

Central Bank of Kenya average lending rates (2025), via HapaKenya and Business Daily; Warehouse Receipt System tariff, via People Daily